

• Menti poll

* avg hours studying per week.

* we might have a bit of time (bif. theory > discuss)

• Matlab S is a bit delayed, but I will post it
tonight.

AND grade past activities until end of day tomorrow.

(solutions will be delayed a bit until everyone submit)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \lambda$$

eigenvector:

$$\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad v_1 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

gen. eigenvector.

$$\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad v_2 = \begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$$

linear system

$$a) \quad A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \quad b) \quad \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} \quad \begin{aligned} \lambda^2 + 6\lambda + 9 &= 0 \\ \lambda &= -3. \end{aligned}$$

$$X(t) = c_1 \cdot v_1 e^{-3t} + c_2 \cdot v_2 t e^{-3t}$$

$$\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 3x + y = 0$$

$$v_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

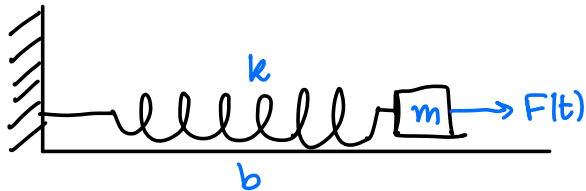
$$v = k \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k \\ 3k \end{bmatrix} \Rightarrow 3x + y = -k \quad v_2 = \begin{bmatrix} 0 \\ -k \end{bmatrix}$$

Lecture 10 Chapter 4. Forcing and Resonance

4.1, 4.2, 4.3, 4.4. (maybe 4.5)

FORCED HARMONIC OSCILLATOR



$$m\ddot{x} + b\dot{x} + kx = F(t)$$

m : mass

b : damping coeff.

k : spring constant

x : position at time t

F : Forcing function

Standard form

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = \frac{F(t)}{m}$$

$$\ddot{x} + p\dot{x} + qx = f(t)$$

$$\ddot{x} + p\dot{x} + qx = 0 \quad \checkmark$$

Extended Linearity Principle.

2nd order DE: $\overbrace{y'' + py' + qy}^{L[y]} = f(t)$

associated homo. eq. $y'' + py' + qy = 0$

then

1. Difference of solutions are solutions for homogeneous part.

$$L[y_1] = f$$

$$L[y_2] = f$$

$$L[\underbrace{y_1 - y_2}_{y_h}] = L[y_1] - L[y_2] = f - f = 0.$$

2. y_p, y_h solutions for non-homo/homo, then $y_p + y_h$ sol. non-homo

$$L[y_p] = f$$

$$L[y_h] = 0$$

$$L[y_p + y_h] = L[y_p] + L[y_h] = f + 0 = f.$$

Solving 2nd order linear non-homogeneous:

Step 1: Find y_h .

Step 2: Find one y_p .

Step 3: Sum it up.

Step 4: Profit.

Qualitative implications ($t \rightarrow \infty$)

$$x'' + px' + qx = 0 \quad \text{with } p, q \geq 0$$

$$y_n \rightarrow 0$$

$$x'' + px' + qx = f$$

$$y = \underbrace{y_n}_{\rightarrow 0} + \underbrace{y_p}_{?} \approx y_p \quad \text{for large } t.$$

$$y(t) = \underbrace{c_1 e^{-2t} + c_2 e^{-3t}}_{y_n \rightarrow 0} + \cos(t) \stackrel{\text{largest}}{\approx} \cos(t)$$

$y_p \approx$ steady-state solution (forced response)

$y_n =$ natural response.

Solving the DE: (a not so distant memory)

Step 1: $\lambda^2 + 5\lambda + 6 = 0 \quad y'' + 5y' + 6y = e^{-t}$
 $(\lambda + 2)(\lambda + 3) = 0$

$$y_n = c_1 e^{-2t} + c_2 e^{-3t}$$

Step 2: Undetermined Coefficients.

$$y_p(t) = Ae^{-t}, \quad y_p' = -Ae^{-t}, \quad y_p'' = Ae^{-t}$$

$$Ae^{-t} + 5(-Ae^{-t}) + 6(Ae^{-t}) = e^{-t} \Rightarrow A - 5A + 6A = 1$$

$$A = 1/2$$

Step 3: $y(t) = c_1 e^{-2t} + c_2 e^{-3t} + \frac{1}{2} e^{-t} \rightarrow 0 \text{ as } t \rightarrow \infty$

poll 1.
 a) $c_1 e^{2t} + c_2 e^{-3t}$ c) $c_1 e^{2t} + c_2 e^{3t}$
 b) $c_1 e^{-2t} + c_2 e^{3t}$ ~~d) $c_1 e^{-2t} + c_2 e^{-3t}$~~

Ex: different forcing function

$$y'' + 5y' + 6y = e^{-2t}$$

Step 1: $y_h = c_1 e^{-2t} + c_2 e^{-3t}$.

Step 2: $y_p = A \cdot e^{-2t}$ } solution.

$$\begin{aligned} y_p(t) &= A t e^{-2t} \\ y_p' &= A e^{-2t} - 2A t e^{-2t} \\ y_p'' &= -2A e^{-2t} - 2A e^{-2t} + 4A t e^{-2t} \\ &= (-4A e^{-2t} + 4A t e^{-2t}) \end{aligned}$$

$$-4A + 4A t + 5(A - 2A t) + 6A t = 1$$

$$t(4A - 10A + 6A) = 0 \cdot t \quad \checkmark$$

$$-4A + 5A = 1$$

$$A = 1.$$

$$y_p(t) = t e^{-2t}.$$

Step 3: $y(t) = c_1 e^{-2t} + c_2 e^{-3t} + t e^{-2t}.$

$$y_p'' + 5y_p' + 6y_p = 0 \neq e^{-2t}.$$

side-note

People were looking for solutions

$$y(t) = \sum_{n=0}^{\infty} a_n \cdot t^n \cdot t$$

$$y(t) = \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} \cdot t = t e^{\lambda t}$$

Ex. that rare case

$$y'' + 4y' + 4y = 2e^{-2t}$$

Step 1 $y_h = c_1 e^{-2t} + c_2 e^{-2t} \cdot t$

Step 2: $y_p = A \cdot t^2 \cdot e^{-2t}$

HW
FILL

Again...

what if $y'' + py' + qy = f + g$?

◦ $y_f = A \cdot [\]$

$$y_p = A[\dots] + B[\dots]$$

◦ $y_g = B \cdot [\]$

$$\mathcal{L}[y_f + y_g] = \mathcal{L}[y_f] + \mathcal{L}[y_g] = f + g$$

$\frac{d}{dt}$ Exponentials are boring, let's do some trig!



Example: $y'' + 2y' + 2y = \sin t$

a) $y_p = A \sin(t) + B \cos(t)$

$$y_p' = A \cos(t) - B \sin(t)$$

$$y_p'' = -A \sin(t) - B \cos(t)$$

$$\lambda^2 + 2\lambda + 2 = 0$$

$$\lambda = \frac{-2 \pm \sqrt{4 - 4 \cdot 2}}{2} = \frac{-2 \pm \sqrt{-4}}{2} = -1 \pm i$$

$$-A \sin(t) - B \cos(t) + 2A \cos(t) - 2B \sin(t) + 2A \sin(t) + 2B \cos(t) = \sin(t)$$

$$-A - 2B + 2A = 1 \Rightarrow A - 2B = 1 \quad A - 2(-2A) = 1 \Rightarrow A = \frac{1}{5}$$

$$-B + 2A + 2B = 0 \Rightarrow B = -2A$$

$$B = -\frac{2}{5}$$

$$y(t) = c_1 e^{-t} \cos(t) + c_2 e^{-t} \sin(t) + \frac{1}{5} \sin(t) - \frac{2}{5} \cos(t)$$

$$y'' + 2y' + 2y = \sin(t)$$

b) Making it more complex: $y_p = a \cdot e^{it}$ (joking, exponentials are back!)

$$y_p' = i a e^{it}, \quad y_p'' = -a e^{it}$$

$$-a e^{it} + 2i a e^{it} + 2a e^{it} = e^{it} a (1 + 2i) = e^{it} = \cos t + i \sin t$$

$$a(1 + 2i) = 1$$

$$a = \frac{1}{1 + 2i} \cdot \frac{(1 - 2i)}{(1 - 2i)} = \frac{1 - 2i}{5}$$

$$\text{Im}(y_p) = \frac{1}{5} \sin t - \frac{2}{5} \cos t$$

break 7:45.

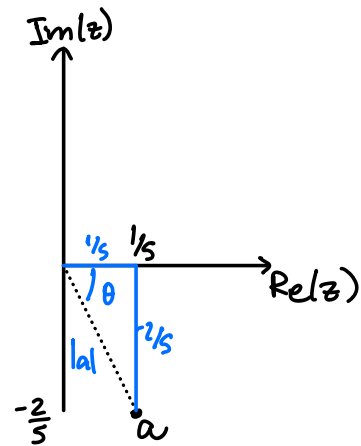
Technicalities:

For $y_c(t) = ae^{it}$,

$$a = |a|e^{i\theta}$$

$$|a| = \sqrt{\left(\frac{1}{5}\right)^2 + \left(\frac{2}{5}\right)^2} = \sqrt{\frac{1+4}{25}} = \frac{1}{\sqrt{5}}$$

$$\theta = \arctan\left(\frac{-2/5}{1/5}\right) = \arctan(-2) \approx 63^\circ$$



$$y_c(t) = \frac{1}{\sqrt{5}} \cdot e^{i\theta} \cdot e^{it} = \frac{1}{\sqrt{5}} e^{i(t+\theta)} = \frac{1}{\sqrt{5}} (\cos(t+\theta) + i \sin(t+\theta))$$

$$\text{Im}(y_c(t)) = \frac{1}{\sqrt{5}} \sin(t+\theta) = \frac{1}{\sqrt{5}} \cos(t+\theta-90^\circ) = \frac{1}{\sqrt{5}} \cos(t-63^\circ)$$

y_p

It is not that complex!

$$A \cos(\omega t) + B \sin(\omega t) = k \cos(\omega t + \phi)$$

$$k = \sqrt{A^2 + B^2}, \quad \phi = \arctan\left(\frac{B}{A}\right)$$

k : amplitude

ω : angular frequency

ϕ : phase

$$x'' + px' + qx = f \quad y = x'$$

$$x' = y$$

$$y' = -qx - py + f$$

Phase Portrait

pros: qualitative analysis / steady-state

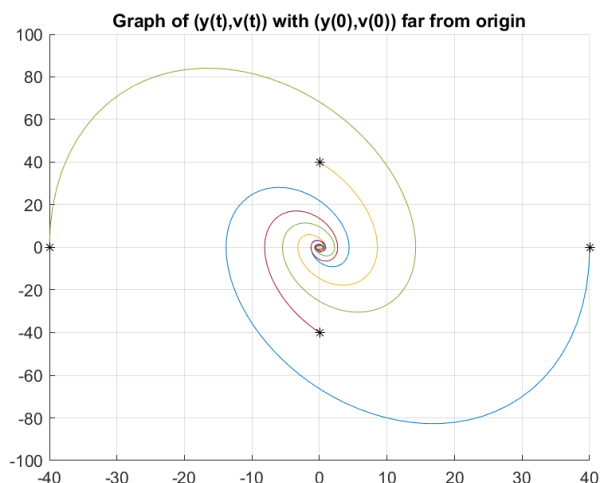
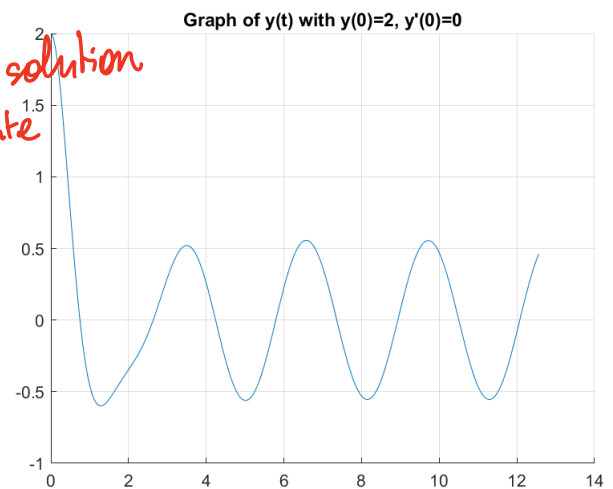
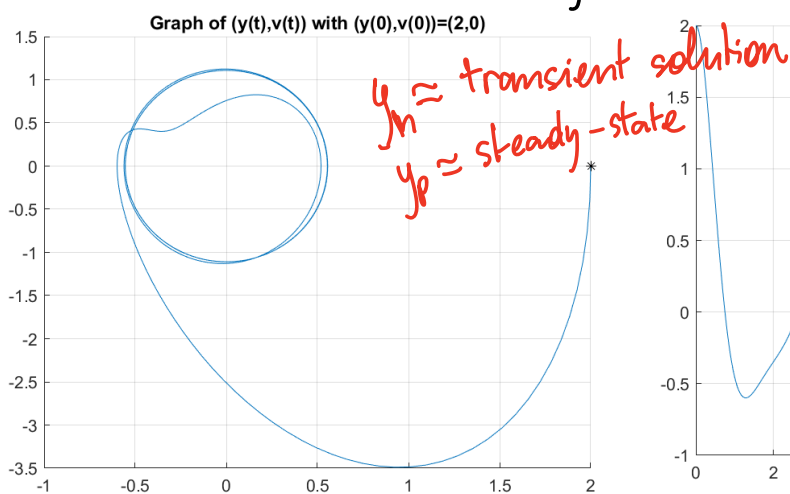
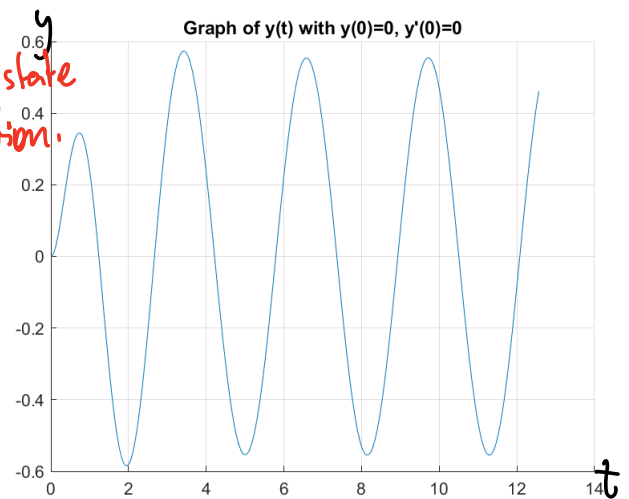
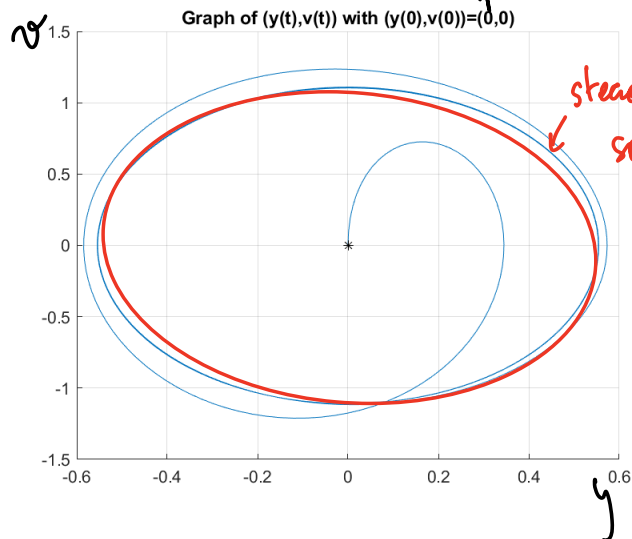
cons: non-autonomous $\Rightarrow$ crossings.

example: $y'' + 2y' + 10y = 4\cos(2t)$

hw Check that $y_p(t) = \underbrace{\frac{6}{13}}_A \cos(2t) + \underbrace{\frac{4}{13}}_B \sin(2t).$

Graphs.

Q: What is the expected behavior?



$$x'' + \rho x' + q x = k \cos(\omega t)$$

Undamped Forcing and Resonance.

Why? If damp is sufficiently small, then this is a good approx. $\ddot{y}$

ex: $\text{Re}\left(\frac{d^2}{dt^2}y + 2y = \cos(\omega t)\right) = \text{Re}(e^{i\omega t})$

$$\text{Re}(e^{i\omega t}) \stackrel{\text{Re}}{=} (\cos(\omega t) + i\sin(\omega t)) \\ = \cos(\omega t)$$

Step 1: $\lambda^2 + 2 = 0 \Rightarrow \lambda = \pm\sqrt{2}i$

$$y_h(t) = c_1 \cos(\sqrt{2}t) + c_2 \sin(\sqrt{2}t)$$

Step 2: $y_p = a \cdot e^{i\omega t}$ (real part)

$$y_p' = ai\omega \cdot e^{i\omega t}$$

$$y_p'' = -a\omega^2 e^{i\omega t}$$

$$(-a\omega^2 e^{i\omega t}) + 2ae^{i\omega t} = e^{i\omega t}$$

$$a(2 - \omega^2) = 1$$

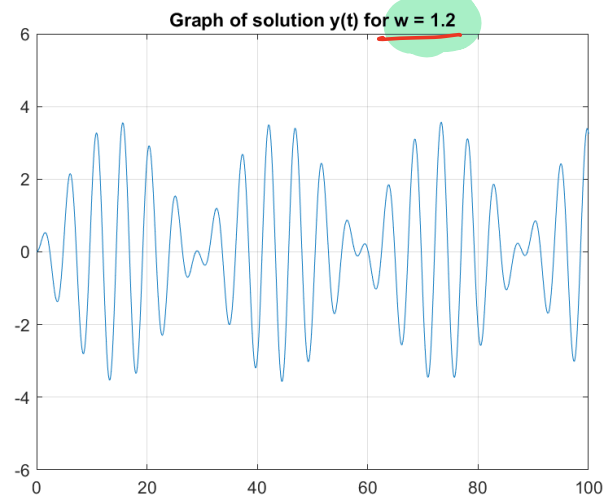
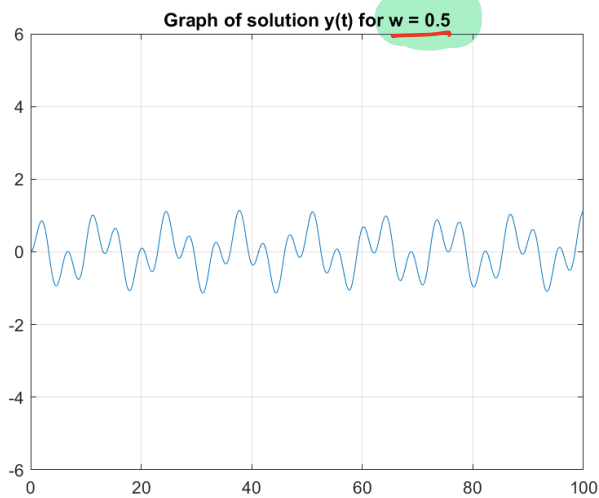
$$a = \frac{1}{2 - \omega^2}$$

$$\text{Re}(y_p) = a \cdot \cos(\omega t) = \frac{1}{2 - \omega^2} \cdot \cos(\omega t)$$

Step 3:

$$y(t) = c_1 \cos(\sqrt{2}t) + c_2 \sin(\sqrt{2}t) + \underbrace{\frac{1}{2 - \omega^2} \cos(\omega t)}_{\text{steady-state}}$$

$$\omega = ?$$



Beating: interference pattern when the natural response and the forced response have approx. the same frequency.

You can hear it too. ;)

(but I don't know how to insert audio here ;))

What is useful for?

tuning instruments



but why is this happening?

gen. sol. $\bullet y(t) = k_1 \cos(\sqrt{2}t) + k_2 \sin(\sqrt{2}t) + a \cos(\omega t)$

For $y(0) = 0, y'(0) = 0$,

$$y(t) = a(\cos(\omega t) - \cos(\sqrt{2}t))$$

Define $\alpha = \frac{\omega + \sqrt{2}}{2} \omega_0$, $\beta = \frac{\omega - \sqrt{2}}{2} \omega_0$

Re $\left\{ \begin{aligned} y_c(t) &= a(e^{i\omega t} - e^{i\sqrt{2}t}) \\ &= a(e^{i(\alpha+\beta)t} - e^{i(\alpha-\beta)t}) \\ &= a e^{i\alpha t} (e^{i\beta t} - e^{-i\beta t}) \\ &= a \cdot e^{i\alpha t} (2i \sin \beta t) \\ &= a \cdot (\cos(\alpha t) + i \sin(\alpha t)) (2i \sin \beta t) \\ &= -2a \sin(\alpha t) \sin(\beta t) + i [a \cos(\alpha t) 2 \sin \beta t] \end{aligned} \right.$

$$\begin{aligned} e^{i\beta t} &= \cos(\beta t) + i \sin(\beta t) \\ e^{-i\beta t} &= \cos(\beta t) - i \sin(\beta t) \end{aligned}$$

$$y(t) = -2a \sin(\alpha t) \sin(\beta t)$$

$\beta/2\pi$: Beating frequency
 $\alpha/2\pi$: wave frequency.

/

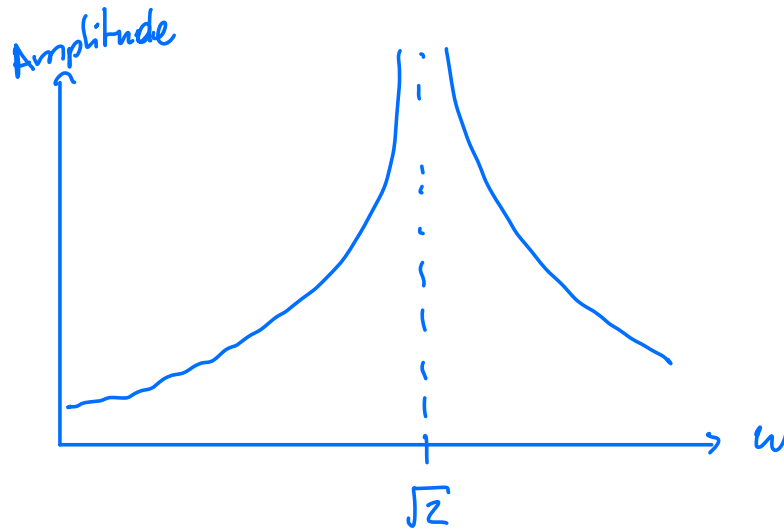
—

Dependence of amplitude on forcing frequency.

• $y'' + 2y = \cos(\omega t)$

$$y(t) = k_1 \cos(\sqrt{2}t) + k_2 \sin(\sqrt{2}t) + \underbrace{\frac{1}{2-\omega^2} \cos(\omega t)}$$

if $\omega \rightarrow 2, \frac{1}{2-\omega^2} \rightarrow \infty$



Resonance

Ex. $y'' + 2y = \cos \omega t$

$\omega = \sqrt{2}$?

Expectation:

- a) solutions go to steady state
- b) solutions oscillate with constant amplitude
- c) solutions go to zero
- d) solutions go to infinity

poll 2

te^{-2t}

Any ideas on how to guess a solution?

$y_p(t) = \underline{A}t \cos(\sqrt{2}t) \leadsto t \cos(\omega t)$ is as big as you want.

$y_h(t) = c_1 \cos(\sqrt{2}t) + c_2 \sin(\sqrt{2}t)$

$y'' + 2y = \cos \sqrt{2}t \sim \operatorname{Re}(y'' + 2y = e^{i\sqrt{2}t})$

$y_p(t) = a t e^{i\sqrt{2}t}$

$y_p' = a e^{i\sqrt{2}t} + (i\sqrt{2}) a t e^{i\sqrt{2}t}$

$y_p'' = (i\sqrt{2}) a e^{i\sqrt{2}t} + a(i\sqrt{2}) e^{i\sqrt{2}t} + (i\sqrt{2})^2 a t e^{i\sqrt{2}t}$

$$-2ate^{i\sqrt{2}t} + 2ai\sqrt{2}e^{i\sqrt{2}t} + 2ate^{i\sqrt{2}t} = e^{i\sqrt{2}t}.$$

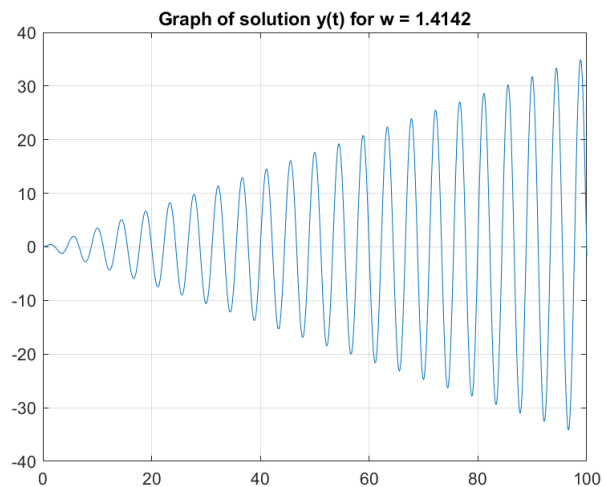
$$(-\cancel{2at} + 2ai\sqrt{2} + \cancel{2at}) = 1.$$

$$a = \frac{1}{2\sqrt{2}i} \cdot \frac{i}{i} = \frac{-i}{2\sqrt{2}}$$

$$\operatorname{Re}\left(\frac{-i}{2\sqrt{2}}te^{i\sqrt{2}t}\right) = \operatorname{Re}\left(\frac{-i}{2\sqrt{2}}t\cos(\sqrt{2}t) - \frac{i}{2\sqrt{2}}t \cdot i\sin(\sqrt{2}t)\right)$$

$$= \frac{t}{2\sqrt{2}}\sin(\sqrt{2}t).$$

Graph for resonance



Real-life problems:



Ex Determine the frequency of the beats, the rapid oscillations and sketch the solution.

$$\underbrace{y'' + 4y}_{\omega_0} = \cos\left(\underbrace{\frac{2t}{4}}_{\omega}\right)$$

$$\frac{\alpha}{2\pi} ? \quad \frac{\beta}{2\pi} ?$$

classical exercise.